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Setu

One pregnancy record, read three different ways — by the mother, by the health worker who visits her, and by the doctor who may never meet her.

A pregnant woman in rural Karnataka is tracked by three people who never share what they know. Her information exists. It simply has no way to travel. Setu is the bridge.

The information exists. It just never travels.

An ASHA worker walks to a pregnant woman's house, checks her blood pressure, and writes the reading into a paper register. If that reading is dangerous, the doctor at the Primary Health Centre finds out when — and if — the woman arrives.

The same pregnancy is being tracked in three disconnected places at once: a Thai Card the mother carries in her bag, a register in the sub-centre, and nothing at all at the PHC. Nobody in this chain is negligent. There is simply no path for the record to move from the doorstep to the doctor while it still matters.

The failure is not detection. It is delivery. Somebody almost always notices in time — and then the observation stops moving.

WHY THIS GAP IS WORTH SOLVING

WHAT IS KNOWN

WHY IT MATTERS HERE

Roughly 15% of pregnancies develop complications

Most are detectable — raised blood pressure, anaemia, poor weight gain — by measurements an ASHA already takes at the door.

High-risk pregnancies account for the large majority of serious maternal and newborn outcomes

The value is almost entirely in identifying *which* pregnancies those are, early, and getting them in front of a clinician.

Studies repeatedly find gaps in frontline referral practice

Not a knowledge failure to be lectured away — a decision-support failure. The threshold should be on the screen, not in memory.

Karnataka has already brought maternal mortality below the SDG target

Which is precisely why the remaining deaths are the hard ones: scattered, rural, and lost in the gap between a home visit and a clinic.

Setu is not aimed at a state that is failing. It is aimed at the last stubborn margin in a state that is otherwise doing well — where the remaining cause is coordination, not care.

Three readers, one shared record

Setu keeps one pregnancy record and shows it three different ways — each shaped for the person reading it and the situation they are standing in. Nobody is asked to maintain a second copy of anything.

THE MOTHER

Thayi Setu

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- Her full pregnancy record, in Kannada, in plain language
- Every checkup, and what each reading actually meant
- Her next visit, and why it matters
- Schemes she qualifies for and what to carry
- An assistant she can ask questions any hour
- Her digital Thayi Card, for the clinic counter

THE HEALTH WORKER

ASHA Setu

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- Today's visits, ordered by who is most at risk
- A visit form that works with no signal at all
- Risk flagged on the spot, before she leaves
- What to say and who to call when it fires
- Follow-up tasks sent down from the doctor
- Her own visit record, ready for her claim

THE MEDICAL OFFICER

Setu Care

- The whole facility ranked by clinical risk
- Each mother's readings plotted over months
- Access only where consent has been given
- Tasks assigned back to the named worker
- Prescriptions captured into the record
- A day's triage list, ready before clinic opens

Same record. Three grammars. The mother reads hers; she never has to maintain it. The worker records; she never has to interpret alone. The doctor triages; he never has to start from a blank page.

The design constraint that produced all three: **assume no network, no second device, and no free time.** Every feature that survived that constraint is in the product. Everything that did not, is not.

The journey of one dangerous reading

This is the entire product, described as the single event it was built for — a blood pressure reading of 165/110 taken in a village house with no mobile signal.

BEFORE**She is registered once, at the start**

The ASHA registers the pregnancy as she does today. The mother receives her Thayi Card with a code on it, and her app is linked to her record. She enters nothing herself.

STEP 1**The visit — with the phone in flight mode**

The worker opens her list, sees who is due, and records the visit at the door: blood pressure, weight, haemoglobin, and any danger signs the woman reports. No signal is required at any point, because the house does not have any.

STEP 2**The check runs on the phone, immediately**

The reading is compared against thresholds *and against her own earlier readings*. A single high number can be an anxious afternoon. A number that is high against five stable months is a warning.

STEP 3**The alert fires in the room, in Kannada**

The screen takes over — what was found, how urgent it is, and what to do now. The worker is not left deciding alone at somebody's doorstep, and the family hears it in their own language while she is still there.

STEP 4**The reading travels the moment there is signal**

Nothing is lost and nothing waits for her to remember. When the phone finds a network — at the sub-centre, on the road, at her own house — the visit joins the shared record on its own.

STEP 5**The doctor's list re-orders itself**

She moves to the top of the facility list before she has arrived, with her trend visible: the five readings that were fine, and the sixth that was not. He is looking at a shape, not a number.

STEP 6**The instruction goes back down to the worker**

He assigns a follow-up to the named ASHA — bring her in, repeat in two days, refer to the taluk hospital. It appears on her phone. The loop closes, which is the part that does not exist today at all.

AFTER**The mother sees it too**

Her app shows the new entry, the appointment, and why it was made — so she is a participant in the decision rather than someone summoned to a clinic without a reason.

Today this loop takes days, or never completes. In Setu it completes within one working day — and the first three steps complete instantly, with no network at all.

Five decisions worth defending

01 Risk is caught at the doorstep, not at the hospital

The warning check runs on the worker's phone with zero connectivity, against the woman's own history rather than a generic table. The alert appears while the person who can act on it is still standing in the room.

Connectivity is not something you can assume in a village. So we assumed it away.

02 The safety net does not depend on AI

Danger words — bleeding, blurred vision, convulsions and others — are caught in Kannada and English before the assistant ever sees the message, and the screen becomes a full escalation. It is deliberately over-sensitive.

A false alarm costs her a phone call. A missed sign costs considerably more.

03 The assistant may only repeat approved answers

Every reply is drawn from a reviewed library of pregnancy answers. The AI rephrases approved material in her language; it is never the source of the medical fact, and it refuses medicine and dosage questions outright.

A wrong answer is corrected by editing one line — not by retraining a model.

04 Consent works the way it works in a real clinic

A doctor cannot open a woman's record by default. He requests access and she approves it — unless she is standing at the counter holding her Thai Card, in which case showing it opens the record temporarily.

Consent modelled as it actually happens, not as a checkbox nobody reads.

05 Guidelines can change without rebuilding the app

The thresholds that decide what counts as dangerous are settings, not something buried inside the software. When a state guideline is revised, it is an edit — and it reaches every phone in the district.

Health policy moves faster than app store releases.

None of these are technical choices. Each one is a decision about what happens to a woman in a village when the system is wrong, slow, or offline — and who pays for it.

The interface is the accessibility argument

The hardest constraint in rural health technology is not the network. It is that the person holding the phone was never the person the software was designed for. Most digital health tools are built to collect data *for administrators*; the worker gets the typing and somebody else gets the insight.

- **Kannada is the default, not a setting.** Nothing is written in English first and translated later.
- **Every text size is one step larger** than a normal app, because she may be reading in direct sunlight with eyesight nobody has ever tested.
- **Nothing important is small enough to mis-tap** on a cracked budget phone with a wet thumb.
- **Red, amber and green never decorate.** Red appears only where something is clinically dangerous, so it keeps its meaning.
- **Everything works without a network** — including the language itself, which renders correctly with no signal.
- **The worker gets something back.** She sees the risk, the trend and the doctor's reply — not just a form to fill.

SAFETY BY DESIGN

RISK	HOW THE DESIGN ANSWERS IT
An AI invents medical advice	It cannot be the source. It may only restate answers a clinician has approved, and it refuses medicines and doses before the question is even sent.
A warning sign is missed	Two independent nets: keyword detection on anything she writes, and threshold checks on every measurement recorded. Both run offline; both are tuned to over-fire.
A woman's record is read without her knowing	Access is closed by default and opened only by her approval or her physical card. Every opening is recorded against a name.
A record is quietly altered	Entries are never overwritten. A correction is a new entry that points at the old one, so the history stays readable and testable.
The worker is blamed for a machine's call	Setu never diagnoses and never overrides. It flags, it suggests the protocol, and it puts the case in front of a clinician faster.

It is a layer, not a replacement

India already has serious maternal health infrastructure. Any honest proposal has to say what exists, what it does not cover, and where a new thing belongs. Setu is deliberately designed to sit *under* these systems and feed them — not to compete with them.

WHAT ALREADY EXISTS	WHAT IT COVERS	THE GAP SETU FILLS
RCH Portal & the 12-digit RCH ID	The national record of every registered pregnancy and the entitlements attached to it.	It is a reporting system, not a working tool at the doorstep. Setu writes into it rather than around it.
ANMOL, the ANM's tablet app	Offline data entry by the ANM, syncing to RCH.	The ANM is not the person in the house. The <i>ASHA</i> is — and she has no tool at all.
Thayi Card / MCP Card	The mother's physical proof of registration and entitlement.	If she loses it, the history is gone. Setu makes it a readable record she keeps, and turns the card into the consent key.
e-PMSMA high-risk tracking	Pays for extra ASHA visits to women identified as high risk.	Identification still depends on judgement and paper. Setu makes the flag automatic and the extra visits evidenced.
ABDM / ABHA health accounts	The national framework for portable, consented health records.	Setu's consent model is built to match it: closed by default, opened by the patient, time-bound, logged.

The one sentence for a health official: **Setu does not create a new database. It closes the loop between the ASHA's doorstep and the medical officer's desk, and hands the result back to the systems you already run.**

Why this is the right moment

The policy intent is already in place — high-risk pregnancies are already meant to be identified early, tracked individually, and escalated. What is missing is the last-mile instrument that makes that intent operational in a village with one bar of signal. Setu is that instrument, and it runs on the phone the worker already carries.

How this reaches society

Maternal health is not a consumer market. Nobody buys this app. It reaches people the only way it can — through the public health system, adopted by a district and paid for as programme spending, not by the woman or the worker.

PHASE 1

One PHC, three months

A single sub-centre cluster — roughly ten ASHA workers and the mothers they already visit. Run alongside the paper register, not instead of it, so nothing is at risk. One question is being answered: does a dangerous reading reach a clinician faster?

PHASE 2

One taluk, under the District Health Society

With the clinical answer library formally signed off, thresholds set to state guidelines, and a proper link into the RCH record so the ASHA stops entering anything twice. Training folded into existing ASHA review meetings.

PHASE 3

District, then state, under NHM

Hassan first, because that is where the work was grounded. Scaling is a training and supervision exercise, not a software one — the apps are already built to run on low-end Android phones with no assumption of connectivity.

WHAT WE WOULD ASK OF GOVERNMENT

THE ASK

WHY IT IS SMALL

One PHC and its ASHA cluster for a pilot

No change to their duties. They do the visits they already do; the phone does the recording they already do on paper.

A medical officer to sign off the answer library

A few hours of review. Content is drawn from WHO and Ministry material; it needs an accountable name against it, not authorship.

Read and write access to RCH for the pilot cohort

This is what removes duplicate entry and makes the tool a help rather than an extra chore.

The official state thresholds

So the alerts match the protocol the district is already accountable to.

Cost: no new hardware, no new staff, no new register. The marginal cost per mother is a few rupees of hosting. The binding constraint on this project is permission and sign-off, not money.

It does not ask anyone to do more work

Karnataka has roughly a lakh ASHA workers already making these visits and already writing these numbers down. Setu does not add a task to their day. It takes the work they already do and makes it visible — the same day — to the one person who can act on it.

For the mother, it turns a card she can lose into a record she can read. For the medical officer, it turns a waiting room into a ranked list, so the woman whose readings have been climbing for three weeks is seen before the one who happened to arrive first.

WHAT SUCCESS WOULD LOOK LIKE

MEASURE	WHY WE CHOSE IT
Hours from a dangerous reading to clinician awareness	The single number this whole project exists to move.
High-risk pregnancies flagged, and how early in gestation	Early is the entire clinical value; late detection is close to no detection.
Share of flagged women who complete the follow-up assigned	Tests whether the loop actually closes, not just whether alerts fire.
Minutes of paperwork saved per ASHA per week	If it costs her time, it will not survive contact with a real workload.

FEASIBILITY

This is not a concept sketch

A working build of all three apps has already been demonstrated on a physical device: a home visit recorded in flight mode with the danger alert firing on the spot, a doctor's console refusing to open a record until consent is granted, and the assistant answering a pregnancy question while declining a medication one. The remaining work is the link between the field and the server — scoped, understood, and small next to what already exists.

Setu ಸೇತು means *bridge*. Not a new record, not a new responsibility for anyone — just a way for what is already known to reach the person who can use it, while it still matters.